# BBD GLUEING: LOCALLY GIVEN DERIVED OBJECTS ON A BASIS GLUE UNIQUELY UNDER NEGATIVE-EXT AND COHOMOLOGICAL-BOUND VANISHING

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the 'small detail omitted' (source line 5705) is the transfer of uniqueness: F is determined by E = a^*F through the bounded-modules equivalence (a^* is fully faithful with quasi-inverse La_* on the bounded range), so the unique-isomorphism uniqueness of E gives that of F - an auxiliary step away from the compatibility construction, which is fully argued; this lemma is the engine invoked by card L46160 (the Cech glueing special case).

*Editorial note.* The label lemma-bbd-glueing names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- The locally-given situation. Same chapter, Situation \label{situation-locally-given}. Line 5629: the data (B, E_U, rho_a) whose solution is to be constructed.
- Hypercoverings adapted to a full subcategory. Hypercoverings chapter, \label{hypercovering-lemma-hypercovering-site}. Lines 5653-5656: a hypercovering L of D with L_n = {U_{n,i}} inside B; its proof's construction of L_0, L_1 is reused at lines 5673-5677.
- Preparation lemma producing the cartesian object on the total site. Same chapter, \label{lemma-prepare-bbd-glueing}. Lines 5657-5660: the cartesian object E on (C/K)_total restricting compatibly to the E_{U_{n,i}}; its compatibility statement is reused at lines 5691-5696.
- Hypercoverings are preserved by morphisms of sites. Hypercoverings chapter, \label{hypercovering-lemma-hypercovering-morphism-sites}. Lines 5662-5663: K = u(L) is a hypercovering.
- Bounded modules equivalence for hypercoverings. Same chapter, \label{lemma-hypercovering-equivalence-bounded-modules}. Lines 5665-5666: E in D^+ is a^*F for some F in D^+(O_C).

*Date*: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0DAY: sources/stacks-chow/tags.txt, source lines 16678–16678. Prerequisite pointer invoked at proof line(s) 5662; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0DAX: sources/stacks-chow/tags.txt, source lines 16677–16677. Prerequisite pointer invoked at proof line(s) 5654; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DA2: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4754–4754. Prerequisite pointer invoked at proof line(s) 5664; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DCA: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5588–5588. Prerequisite pointer invoked at proof line(s) 5657; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0D67: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5552–5552. Prerequisite pointer invoked at proof line(s) 5631; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1** (BBD glueing lemma)**.** *In Situation 0D67. Assume*

(1) $\mathcal{C}$ *has equalizers and fibre products,*
(2) *there is a morphism of sites* $f : \mathcal{C} \to \mathcal{D}$ *given by a continuous functor* $u : \mathcal{D} \to \mathcal{C}$ *such that*
   (a) $\mathcal{D}$ *has equalizers and fibre products and* $u$ *commutes with them,*
   (b) $\mathcal{B}$ *is a full subcategory of* $\mathcal{D}$ *and* $u : \mathcal{B} \to \mathcal{C}$ *is the restriction of* $u$*,*
   (c) *every object of* $\mathcal{D}$ *has a covering whose members are objects of* $\mathcal{B}$*,*
(3) *for all* $U$ *in* $\mathcal{B}$ *all negative self-exts of* $E_U$ *in* $D(\mathcal{O}_{u(U)})$ *are zero, and*
(4) *there exists a* $t \in \mathbf{Z}$ *such that* $H^i(E_U) = 0$ *for* $i < t$ *and* $U \in \mathrm{Ob}(\mathcal{B})$*.*

*Then there exists a solution unique up to unique isomorphism.*

*Proof.* By Hypercoverings, Lemma 0DAX there exists a hypercovering $L$ for the site $\mathcal{D}$ such that $L_n = \{U_{n,i}\}_{i \in I_n}$ with $U_{i,n} \in \mathrm{Ob}(\mathcal{B})$. Set $K = u(L)$. Apply Lemma 0DCA to get a cartesian object $E$ of $D(\mathcal{O})$ on the site $(\mathcal{C}/K)_{total}$ restricting to $E_{U_{n,i}}$ on $\mathcal{C}/u(U_{n,i})$ compatibly. The assumption on $t$ implies that $E \in D^+(\mathcal{O})$. By Hypercoverings, Lemma 0DAY we see that $K$ is a hypercovering too. By Lemma 0DA2 we find that $E = a^*F$ for some $F$ in $D^+(\mathcal{O}_\mathcal{C})$.

To prove that $F$ is a solution we will use the construction of $L_0$ and $L_1$ given in the proof of Hypercoverings, Lemma 0DAX. (This is a bit inelegant but there does not seem to be a completely straightforward way around it.)

Namely, we have $I_0 = \mathrm{Ob}(\mathcal{B})$ and so $L_0 = \{U\}_{U \in \mathrm{Ob}(\mathcal{B})}$. Hence the isomorphism $a^*F \to E$ restricted to the components $\mathcal{C}/u(U)$ of $\mathcal{C}/K_0$ defines isomorphisms $\rho_U : F|_{\mathcal{C}/u(U)} \to E_U$ for $U \in \mathrm{Ob}(\mathcal{B})$ by our choice of $E$.

To prove that $\rho_U$ satisfy the requirement of compatibility with the maps $\rho_a$ of Situation 0D67 we use that $I_1$ contains the set

$$\Omega = \{(U, V, W, a, b) \mid U, V, W \in \mathcal{B}, a : U \to V, b : U \to W\}$$

and that for $i = (U, V, W, a, b)$ in $\Omega$ we have $U_{1,i} = U$. Moreover, the component maps $f_{\delta^1_0,i}$ and $f_{\delta^1_1,i}$ of the two morphisms $K_1 \to K_0$ are the morphisms

$$a : U \to V \quad \text{and} \quad b : U \to V$$

Hence the compatibility mentioned in Lemma 0DCA gives that

$$\rho_a \circ \rho_V|_{\mathcal{C}/u(U)} = \rho_U \quad \text{and} \quad \rho_b \circ \rho_W|_{\mathcal{C}/u(U)} = \rho_U$$

Taking $i = (U, V, U, a, \mathrm{id}_U) \in \Omega$ for example, we find that we have the desired compatibility. The uniqueness of $F$ follows from the uniqueness of $E$ in the previous lemma (small detail omitted). □